

Module 4 — Electrons, Waves and Photons

This module covers electricity, wave behaviour, and quantum physics — bridging classical and modern physics.

4.1 Charge, Current, and Potential Difference

Electric current (I): The rate of flow of electric charge. Conventional current flows from positive to negative (opposite to electron flow).

Charge (Q): A fundamental property of matter. The elementary charge $e = 1.60 \times 10^{-19} \text{ C}$.

$$Q = It$$

Where Q = charge (C), I = current (A), t = time (s). One coulomb is the charge carried by a current of 1 A in 1 s.

Potential difference (V): The energy transferred per unit charge between two points. $1 \text{ V} = 1 \text{ J C}^{-1}$.

$$V = W / Q$$

Where V = potential difference (V), W = energy transferred (J), Q = charge (C).

Electron volt (eV): The energy gained by an electron accelerated through a PD of 1 V.

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

4.2 Resistance and Ohm's Law

Resistance (R): The opposition to current flow. Defined as the ratio of PD across a component to the current through it.

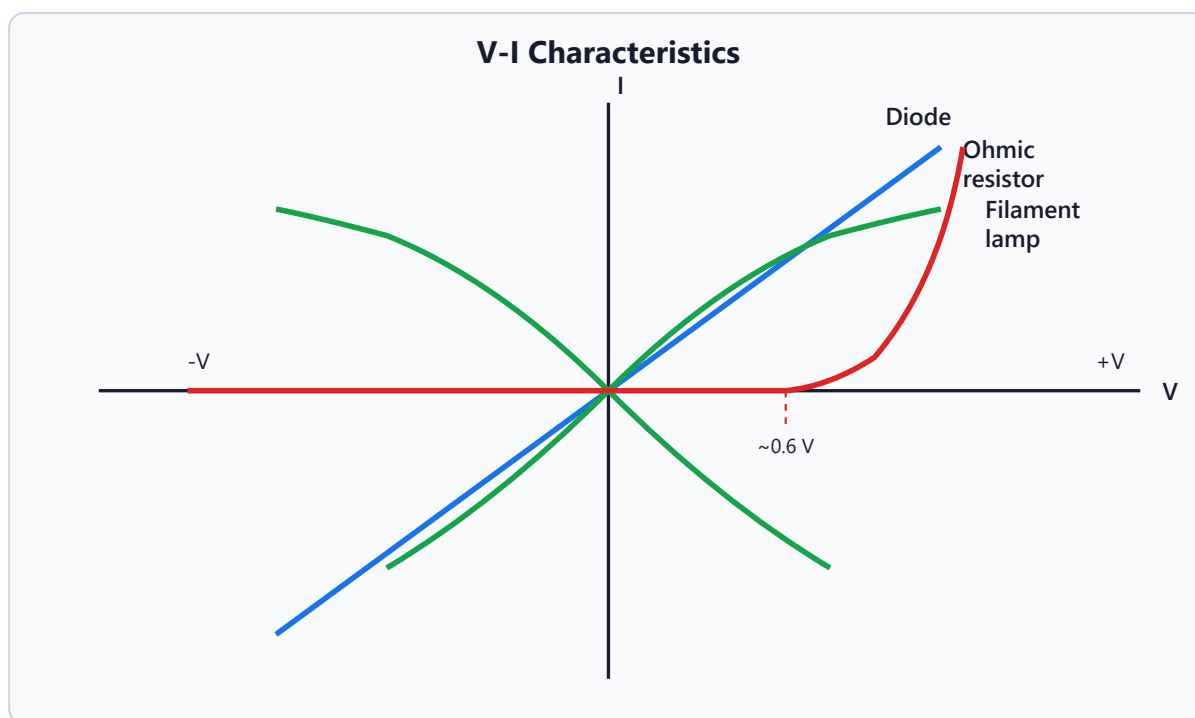
$$V = IR \Rightarrow R = V / I$$

Units: ohms (Ω), where $1 \Omega = 1 \text{ V A}^{-1}$.

Ohm's Law: The current through a conductor is directly proportional to the PD across it, provided physical conditions (especially temperature) remain constant. A component that obeys this is called an ohmic conductor.

I-V Characteristics

Component	I-V Graph Shape	Behaviour
Ohmic resistor (wire)	Straight line through the origin	Constant resistance. Current proportional to PD. Same in both directions.
Filament lamp	S-shaped curve through the origin	Resistance increases with current (and temperature). As current increases, the filament heats up, atoms vibrate more, and electrons collide more frequently with the lattice.
Semiconductor diode	Negligible current for negative PD; exponential increase above $\sim 0.6 \text{ V}$ forward bias	Very high resistance in reverse bias; low resistance in forward bias above the threshold voltage ($\sim 0.6 \text{ V}$ for silicon).
Thermistor (NTC)	Non-linear	Resistance decreases as temperature increases. More charge carriers are released at higher temperatures.
LDR	Non-linear	Resistance decreases as light intensity increases. More photons release more charge carriers in the semiconductor.



Exam Tip: For I-V characteristics, always state whether the component is ohmic or non-ohmic. An ohmic component has a straight line through the origin. For a filament lamp, the curve gets less steep at higher currents because resistance is increasing.

4.3 Resistivity

Resistivity (ρ): A material property that quantifies how strongly it resists current flow, independent of the dimensions of the sample.

$$\rho = RA / L$$

Where ρ = resistivity ($\Omega \text{ m}$), R = resistance (Ω), A = cross-sectional area (m^2), L = length (m).

- Metals: low resistivity ($\sim 10^{-8} \Omega \text{ m}$), good conductors
- Semiconductors: intermediate resistivity ($\sim 10^{-1}$ to $10^3 \Omega \text{ m}$)
- Insulators: very high resistivity ($\sim 10^{10}$ to $10^{16} \Omega \text{ m}$)

4.4 Circuit Rules — Kirchhoff's Laws

Kirchhoff's First Law (junction rule): The sum of currents entering a junction equals the sum of currents leaving it. This is based on conservation of charge.

$$\Sigma I_{\text{in}} = \Sigma I_{\text{out}}$$

Kirchhoff's Second Law (loop rule): The sum of EMFs in any closed loop equals the sum of PDs around that loop. This is based on conservation of energy.

$$\Sigma \mathcal{E} = \Sigma IR$$

Series Circuits

- Current is the same through all components: $I_{\text{total}} = I_1 = I_2 = \dots$
- PDs add up: $V_{\text{total}} = V_1 + V_2 + \dots$
- Resistances add up: $R_{\text{total}} = R_1 + R_2 + \dots$

Parallel Circuits

- PD is the same across all branches: $V_{\text{total}} = V_1 = V_2 = \dots$
- Currents add up: $I_{\text{total}} = I_1 + I_2 + \dots$
- Reciprocals of resistance add: $1/R_{\text{total}} = 1/R_1 + 1/R_2 + \dots$

4.5 EMF and Internal Resistance

Electromotive force (EMF, $\mathcal{E}$): The energy transferred per unit charge by the source (battery/cell). It is the total PD when no current flows.

Internal resistance (r): The resistance within the source itself, which causes energy to be dissipated as heat inside the source.

$$\mathcal{E} = I(R + r) = IR + Ir = V + Ir$$

Where ϵ = EMF (V), I = current (A), R = external resistance (Ω), r = internal resistance (Ω), V = terminal PD (V).

- The terminal PD ($V = IR$) is always less than the EMF when current flows, because some energy is lost in the internal resistance
- "Lost volts" = Ir (the PD dropped across the internal resistance)
- To find ϵ and r experimentally: plot V against I . The y-intercept = ϵ , the gradient = $-r$

Exam Tip: The EMF can be measured by connecting a high-resistance voltmeter across the terminals with no external circuit (open circuit), so $I \approx 0$ and $V \approx \epsilon$.

4.6 Potential Dividers

Potential divider: A circuit with two (or more) resistors in series, used to produce a variable output voltage from a fixed supply.

$$V_{\text{out}} = V_{\text{in}} \times R_2 / (R_1 + R_2)$$

Applications:

- **With a thermistor:** As temperature rises, thermistor resistance falls, changing V_{out} . Used in temperature-sensing circuits.
- **With an LDR:** As light intensity increases, LDR resistance falls, changing V_{out} . Used in light-sensing circuits.
- **With a potentiometer:** A variable resistor used to give a continuously variable output voltage from 0 to V_{in} .

4.7 Electrical Power and Energy

$$P = IV = I^2R = V^2 / R$$

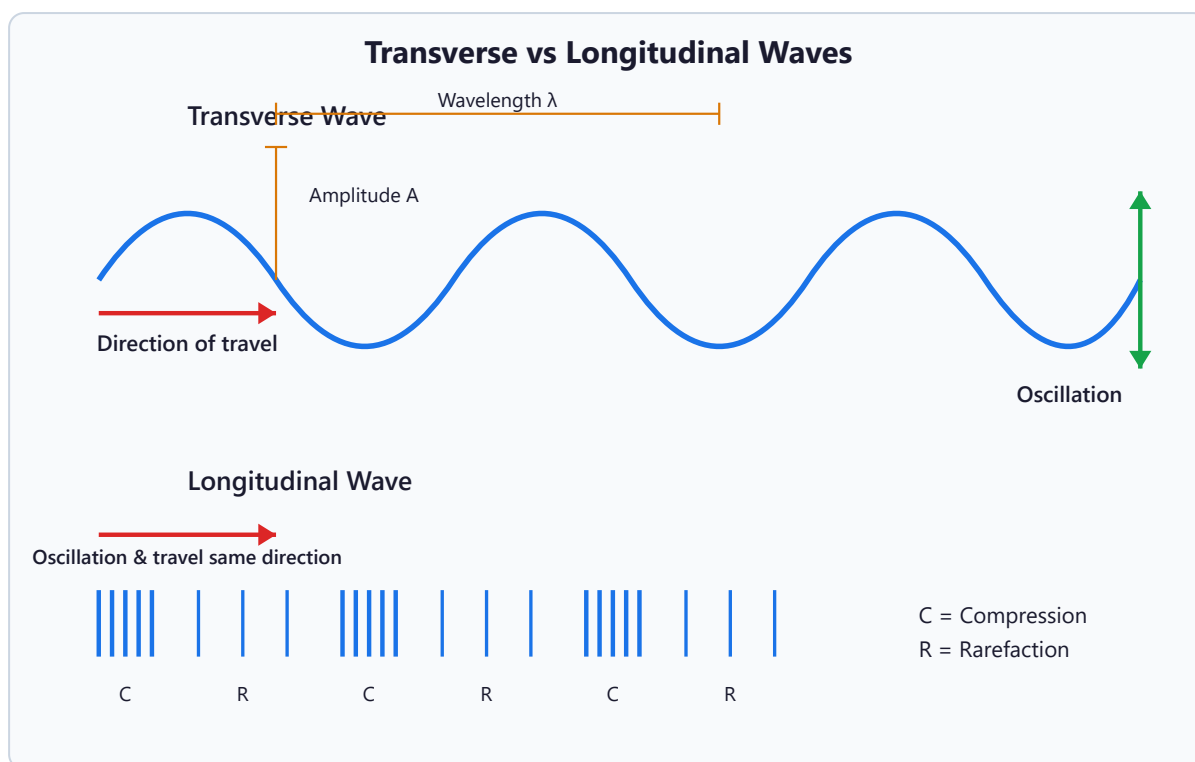
$$E = Pt = IVt = I^2Rt$$

Units: power in watts (W), energy in joules (J).

4.8 Progressive Waves

Progressive wave: A wave that transfers energy from one place to another without transferring matter. Particles oscillate about their equilibrium positions.

Property	Transverse Wave	Longitudinal Wave
Oscillation direction	Perpendicular to direction of energy transfer	Parallel to direction of energy transfer
Examples	EM waves, water waves, waves on a string, S-waves	Sound waves, ultrasound, P-waves, compression springs
Can be polarised?	Yes	No
Features	Crests and troughs	Compressions and rarefactions



Wave Quantities

- **Wavelength (λ):** Distance between two consecutive points in phase (e.g., crest to crest). Units: m
- **Frequency (f):** Number of complete oscillations per second. Units: $\text{Hz (s}^{-1}\text{)}$
- **Period (T):** Time for one complete oscillation. $T = 1/f$. Units: s
- **Amplitude (A):** Maximum displacement from the equilibrium position. Units: m
- **Phase difference:** The fraction of a cycle between two points, measured in degrees or radians. In phase = $0^\circ/360^\circ$; antiphase = 180°

The Wave Equation

$$v = f\lambda$$

Where v = wave speed (m s^{-1}), f = frequency (Hz), λ = wavelength (m).

4.9 Superposition and Stationary Waves

Principle of superposition: When two or more waves meet at a point, the resultant displacement is

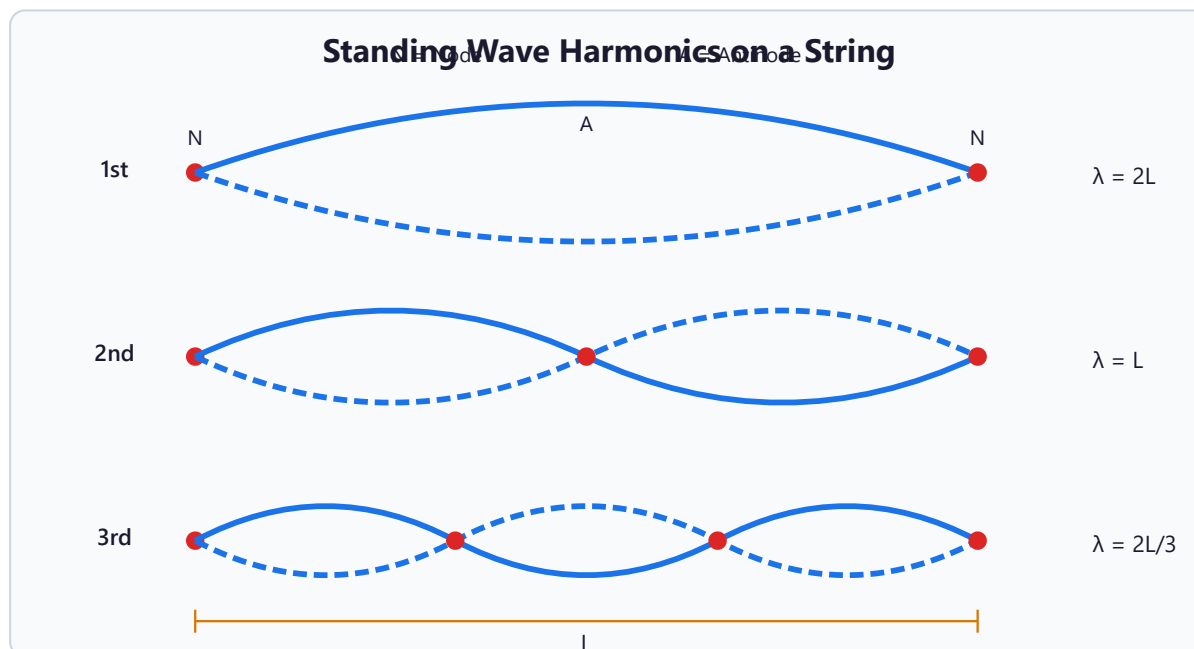
the vector sum of the individual displacements.

- **Constructive interference:** Waves in phase; displacements add. Resultant amplitude is maximised.
- **Destructive interference:** Waves in antiphase; displacements cancel. Resultant amplitude is minimised (or zero for equal amplitudes).

Stationary (Standing) Waves

Stationary wave: Formed by the superposition of two progressive waves of the same frequency, amplitude, and speed, travelling in opposite directions.

- **Nodes:** Points of zero displacement (permanent destructive interference). Separated by $\lambda/2$.
- **Antinodes:** Points of maximum displacement (constructive interference). Midway between nodes.
- No energy is transferred along a stationary wave
- All points between two adjacent nodes oscillate in phase; points on opposite sides of a node oscillate in antiphase



Harmonics on a String

Harmonic	Nodes	Antinodes	Wavelength	Frequency
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1st (fundamental)	2	1	$\lambda = 2L$	f_0
2nd	3	2	$\lambda = L$	$2f_0$
3rd	4	3	$\lambda = 2L/3$	$3f_0$
nth	$n + 1$	n	$\lambda = 2L/n$	nf_0

For a pipe open at both ends: antinodes at both ends (same harmonics as a string).

For a pipe closed at one end: node at closed end, antinode at open end. Only odd harmonics (1st, 3rd, 5th...) are possible. Fundamental: $\lambda = 4L$.

4.10 Refraction

Refraction: The change in direction of a wave when it passes from one medium to another, caused by a change in wave speed. Frequency remains constant; wavelength and speed change.

Snell's Law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Where n = refractive index, θ = angle to the normal.

$$n = c / v$$

Where c = speed of light in a vacuum ($3.00 \times 10^8 \text{ m s}^{-1}$), v = speed of light in the medium.

Total Internal Reflection

Total internal reflection (TIR): Occurs when light travels from a denser medium to a less dense medium, and the angle of incidence exceeds the critical angle. All light is reflected back into the denser medium.

Conditions for TIR: (1) Light must travel from a medium with higher n to lower n . (2) Angle of incidence must exceed the critical angle.

$$\sin \theta_c = n_2 / n_1$$

Applications: optical fibres, endoscopes, prisms in binoculars.

4.11 Diffraction and Interference

Diffraction: The spreading of a wave as it passes through a gap or around an obstacle. Maximum diffraction occurs when the gap width is approximately equal to the wavelength.

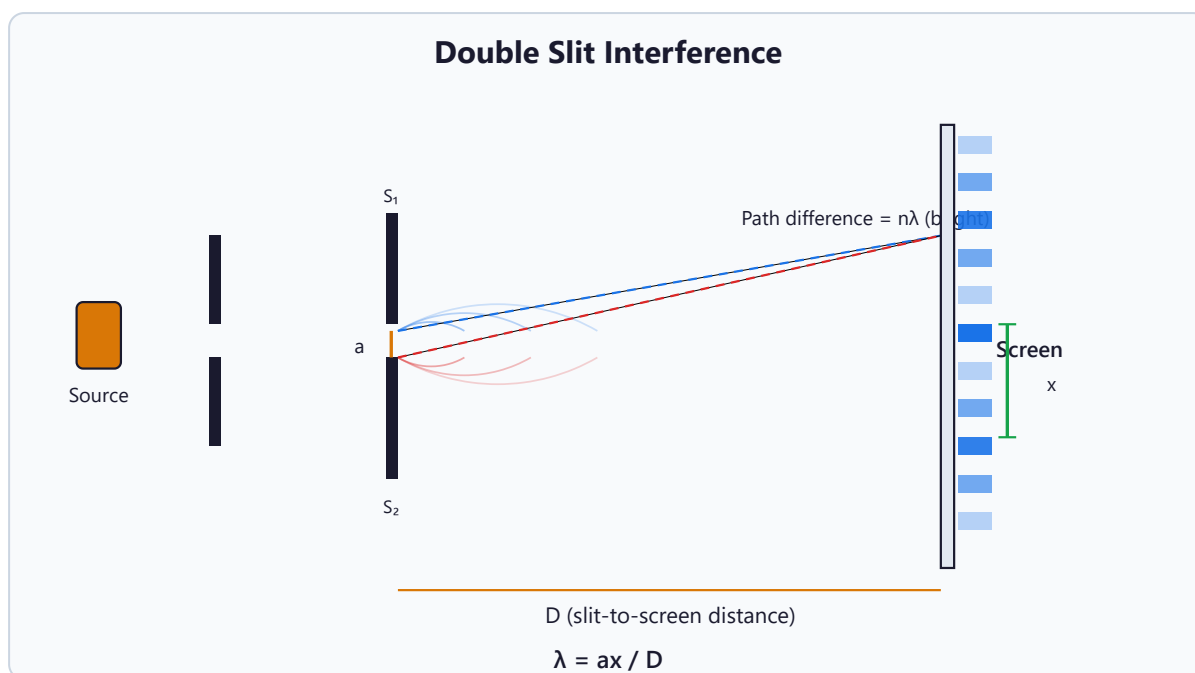
Young's Double Slit Experiment

Coherent light passes through two narrow slits. An interference pattern of bright and dark fringes is observed on a screen.

- **Bright fringes (maxima):** Constructive interference; path difference = $n\lambda$ (where $n = 0, 1, 2, \dots$)
- **Dark fringes (minima):** Destructive interference; path difference = $(n + \frac{1}{2})\lambda$

$$\lambda = ax / D$$

Where a = slit separation (m), x = fringe spacing (m), D = distance from slits to screen (m).



Exam Tip: For interference to be observable, the sources must be coherent (same frequency, constant phase difference). Young's experiment demonstrates that light is a wave — this cannot be explained by a particle model.

Diffraction Gratings

$$d \sin \theta = n\lambda$$

Where d = slit spacing (m), θ = angle of diffraction, n = order number (0, 1, 2, ...), λ = wavelength (m).

- $d = 1 / N$, where N = number of slits per metre
- Higher orders appear at larger angles
- Maximum order: $n_{\text{max}} = d / \lambda$ (rounded down, since $\sin \theta \leq 1$)
- Gratings produce sharper, brighter maxima than double slits
- White light produces a spectrum at each order (violet closest to centre, red furthest)

4.12 Polarisation

Polarisation: The restriction of the oscillations of a transverse wave to a single plane. Only transverse waves can be polarised — this is evidence that EM waves are transverse.

- Unpolarised light oscillates in all planes perpendicular to the direction of travel
- A polarising filter (Polaroid) transmits only oscillations in one plane
- Two crossed polarisers (at 90°) block all light — Malus's Law: $I = I_0 \cos^2\theta$
- Applications: polarising sunglasses (reduce glare), LCD screens, stress analysis

4.13 Electromagnetic Spectrum

Type	Wavelength Range	Source Examples
Radio waves	$> 0.1 \text{ m}$	Oscillating circuits, transmitters
Microwaves	$0.1 \text{ m} - 1 \text{ mm}$	Magnetron, cosmic background
Infrared	$1 \text{ mm} - 700 \text{ nm}$	Hot objects, heaters
Visible light	$700 \text{ nm} - 400 \text{ nm}$	Sun, lamps, lasers
Ultraviolet	$400 \text{ nm} - 10 \text{ nm}$	Sun, UV lamps, very hot objects
X-rays	$10 \text{ nm} - 0.01 \text{ nm}$	X-ray tubes, synchrotrons
Gamma rays	$< 0.01 \text{ nm}$	Radioactive decay, nuclear reactions

All EM waves travel at $c = 3.00 \times 10^8 \text{ m s}^{-1}$ in a vacuum, are transverse, and can be polarised.

4.14 Photon Model and the Photoelectric Effect

Photon: A quantum (discrete packet) of electromagnetic energy. The energy of a photon is proportional to its frequency.

$$E = hf = hc / \lambda$$

Where h = Planck's constant = $6.63 \times 10^{-34} \text{ J s}$.

The Photoelectric Effect

Photoelectric effect: The emission of electrons from the surface of a metal when electromagnetic radiation of sufficiently high frequency is incident on it.

Key observations that cannot be explained by the wave model:

1. Electrons are emitted instantaneously (no time delay), even at low intensity
2. Below the threshold frequency f_0 , no electrons are emitted regardless of intensity
3. Increasing intensity increases the number of photoelectrons, not their maximum kinetic energy
4. Increasing frequency above f_0 increases the maximum kinetic energy of photoelectrons

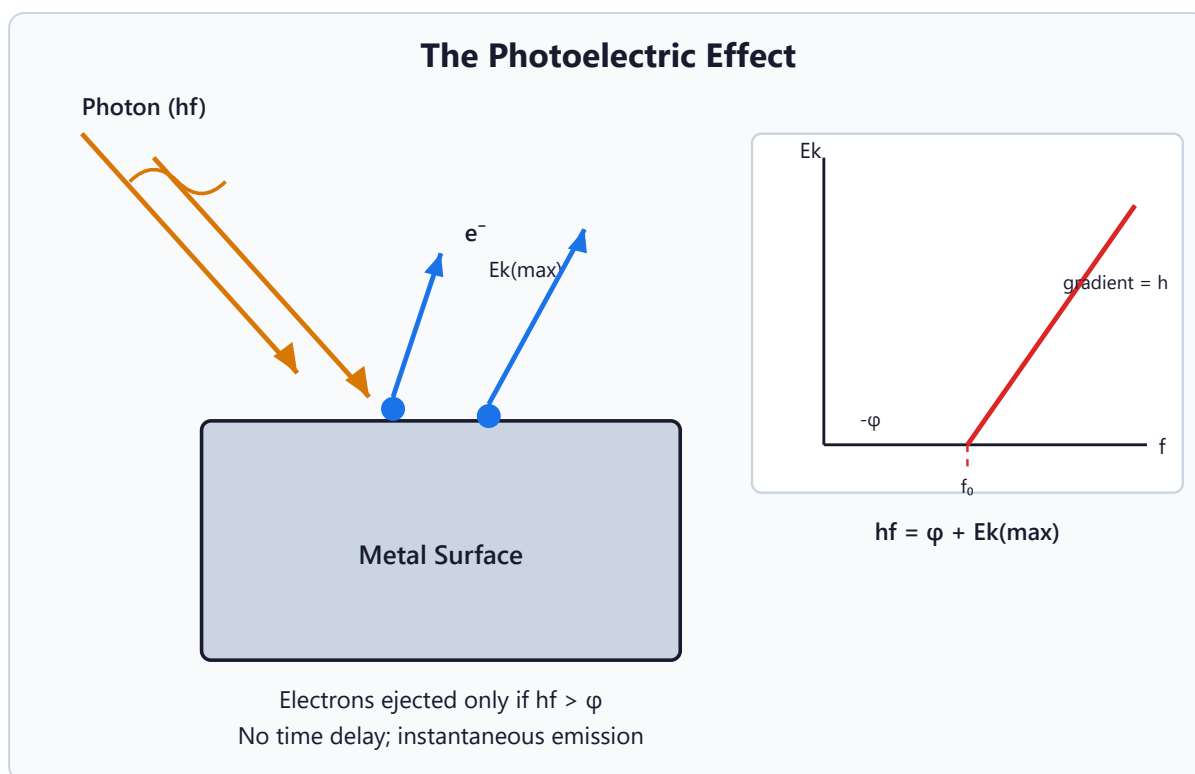
Einstein's Photoelectric Equation

$$hf = \phi + E_{k(\max)}$$

Where hf = energy of the incident photon, ϕ = work function (minimum energy to release an electron from the surface), $E_{k(\max)}$ = maximum kinetic energy of the emitted electron.

Threshold frequency (f_0): The minimum frequency of radiation needed to release electrons. At the threshold: $hf_0 = \phi$, so $f_0 = \phi / h$.

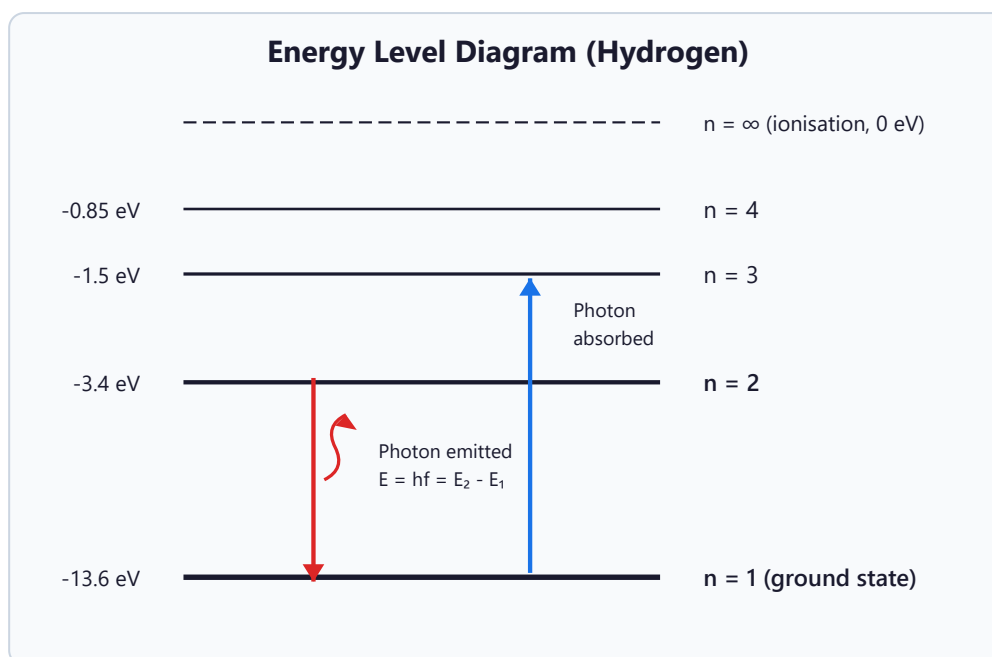
Work function (ϕ): The minimum energy needed to free an electron from the metal surface.



Exam Tip: On a graph of $E_{k(max)}$ vs frequency, the gradient = h (Planck's constant), the y-intercept = $-\phi$, and the x-intercept = f_0 (threshold frequency). Different metals have different work functions but the same gradient.

4.15 Energy Levels and Line Spectra

Energy levels: Electrons in atoms can only exist at specific, discrete energy values. The ground state is the lowest energy level; excited states are higher.



- When an electron moves from a higher energy level E_2 to a lower level E_1 , a photon is emitted with energy: $E = hf = E_2 - E_1$
- When a photon with exactly the right energy is absorbed, an electron moves up to a higher level
- **Emission spectrum:** Bright coloured lines on a dark background (from excited gas). Each line corresponds to a specific transition between energy levels.
- **Absorption spectrum:** Dark lines on a continuous coloured background. The dark lines correspond to photons absorbed by the gas at specific frequencies.

4.16 De Broglie Wavelength

Wave-particle duality: All matter exhibits both wave and particle properties. The de Broglie wavelength is the wavelength associated with a moving particle.

$$\lambda = h / mv = h / p$$

Where λ = de Broglie wavelength (m), h = Planck's constant, m = mass (kg), v = velocity (m s^{-1}), p = momentum (kg m s^{-1}).

- Electrons show wave behaviour (diffraction) when their de Broglie wavelength is comparable to atomic spacing ($\sim 10^{-10}$ m)

- Evidence: electron diffraction patterns (rings) when electrons pass through graphite
- Faster electrons have shorter wavelengths; more massive particles have shorter wavelengths

Exam Tip: The de Broglie wavelength applies to all particles, not just electrons. However, for macroscopic objects the wavelength is negligibly small and wave effects are undetectable.

Worked Example: Calculate the de Broglie wavelength of an electron accelerated through 200 V.
($m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C, $h = 6.63 \times 10^{-34}$ J s)

$$\text{KE gained} = eV = 1.60 \times 10^{-19} \times 200 = 3.20 \times 10^{-17} \text{ J}$$

$$\frac{1}{2}mv^2 = 3.20 \times 10^{-17}, \text{ so } v = \sqrt{2 \times 3.20 \times 10^{-17} / 9.11 \times 10^{-31}} = 8.38 \times 10^6 \text{ m s}^{-1}$$

$$\lambda = h/mv = 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 8.38 \times 10^6) = \mathbf{8.7 \times 10^{-11} \text{ m} = 0.087 \text{ nm}}$$

This is comparable to atomic spacing, so electron diffraction can be observed when such electrons pass through a crystal lattice.

Worked Example: A hydrogen atom transitions from $n = 3$ ($E = -1.5$ eV) to $n = 2$ ($E = -3.4$ eV). Calculate the wavelength of the emitted photon.

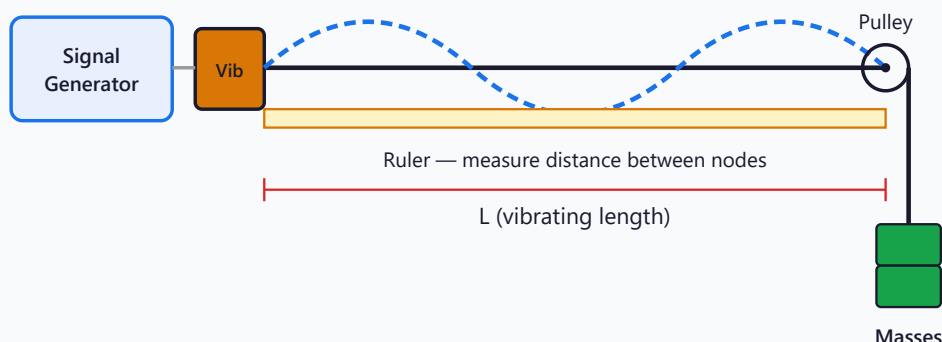
$$\text{Energy of photon} = E_3 - E_2 = -1.5 - (-3.4) = 1.9 \text{ eV} = 1.9 \times 1.60 \times 10^{-19} = 3.04 \times 10^{-19} \text{ J}$$

$$E = hf = hc/\lambda, \text{ so } \lambda = hc/E = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / 3.04 \times 10^{-19} = \mathbf{6.54 \times 10^{-7} \text{ m} = 654 \text{ nm}}$$

This is red light — it is the H-alpha line in the hydrogen emission spectrum, one of the most important spectral lines in astronomy for identifying hydrogen in stars and nebulae.

PAG 5: Stationary Waves on a String

PAG 5 — Stationary Waves Apparatus



Apparatus

- Signal generator and vibration generator (mechanical oscillator)
- String (thin, low mass per unit length)
- Pulley clamped to the bench edge
- Slotted masses and hanger to provide tension
- Metre ruler
- Strobe light (optional, to freeze the wave pattern)

Method

1. Set up the string between the vibrator and pulley, with masses hanging over the pulley to provide tension $T = mg$
2. Adjust the frequency on the signal generator until a clear standing wave pattern is visible (start with the fundamental)
3. Measure the distance between adjacent nodes using the ruler to find $\lambda/2$
4. Record the frequency f from the signal generator and calculate $\lambda = 2 \times (\text{node separation})$
5. Calculate the wave speed: $v = f\lambda$
6. Repeat for different harmonics (2nd, 3rd, etc.) by increasing frequency
7. To investigate the effect of tension, keep frequency constant and vary the hanging mass
8. Verify that $v = \sqrt{T/\mu}$, where μ = mass per unit length of string

Safety

- Wear safety goggles in case the string snaps
- Place a cushion below the hanging masses
- If using a strobe, warn about photosensitive epilepsy

Analysis

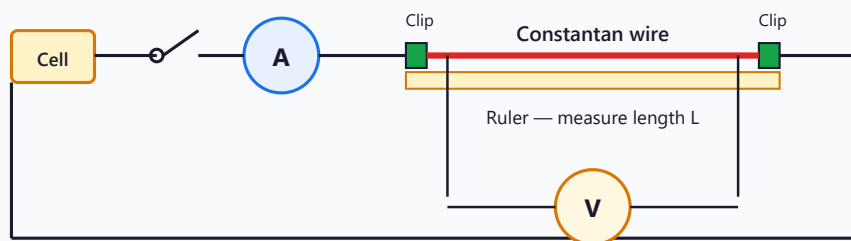
Plot f against $1/\lambda$ — the gradient equals the wave speed v . Alternatively, plot v^2 against T — the gradient equals $1/\mu$ (mass per unit length).

Sources of Error & Improvements

Source of Error	Type	Improvement
Difficulty judging exact node positions	Random	Use a paper rider — it falls off at a node; use a strobe light to freeze the pattern
String may not vibrate in a clean single harmonic	Systematic	Fine-tune frequency carefully; increase amplitude slowly
Mass of hanging string section not accounted for	Systematic	Use a long vibrating section so the hanging section is a small fraction

Exam Tip: The fundamental frequency of a string depends on its length, tension, and mass per unit length: $f = (1/2L)\sqrt{T/\mu}$. Increasing tension or decreasing length increases the fundamental frequency.

PAG 3: Resistivity of a Wire

PAG 3 — Resistivity Apparatus

Also measure diameter d with micrometer

$$\rho = RA/L \text{ where } A = \pi d^2/4$$

Apparatus

- Constantan wire (SWG 28 or 30) mounted on a ruler
- Crocodile clips to connect at different lengths along the wire
- Ammeter in series, voltmeter in parallel across the wire
- Cell or power supply (low voltage, e.g. 1.5 V) and switch
- Micrometer screw gauge to measure wire diameter
- Metre ruler

Method

1. Measure the diameter d of the wire at several points using the micrometer; calculate mean diameter and cross-sectional area $A = \pi d^2/4$
2. Connect the circuit with the ammeter in series and voltmeter across the test length of wire
3. Set the crocodile clips to the first length (e.g. $L = 0.10 \text{ m}$)
4. Close the switch briefly, record V and I , calculate $R = V/I$
5. Repeat for at least 6 different lengths up to $\sim 1.0 \text{ m}$
6. Plot R (y-axis) against L (x-axis)
7. Gradient = ρ/A , so $\rho = \text{gradient} \times A$

Safety

- Keep current low and switch off between readings to prevent the wire heating up (which would change its resistance)

- Do not touch the wire while current is flowing

Analysis

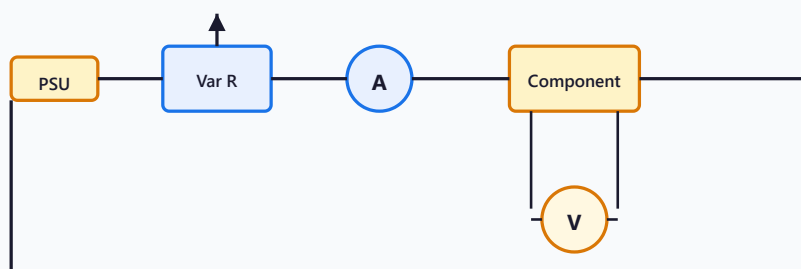
From $\rho = RA/L$, rearranging gives $R = (\rho/A)L$. A graph of R vs L is a straight line through the origin with gradient $= \rho/A$.

Sources of Error & Improvements

Source of Error	Type	Improvement
Wire heats up during measurement, changing resistance	Systematic	Use low current; switch off between readings; use a switch to keep circuit off
Difficulty measuring exact length between clips	Random	Use sharp crocodile clips positioned precisely; repeat readings
Diameter varies along the wire	Random	Measure diameter at 6+ points and different orientations; use the mean
Contact resistance at crocodile clips	Systematic	Clean contacts; use voltmeter across wire (not across clips + wire)

Exam Tip: The voltmeter must be connected across the wire only (between the clips), not across the whole circuit, otherwise it measures the EMF, not the PD across the wire.

PAG 3b: V-I Characteristics

PAG 3b — V-I Characteristics Circuit

Test: filament lamp, resistor, diode (reverse connections for diode)

Apparatus

- DC power supply (variable voltage) or cell with variable resistor (rheostat)
- Ammeter in series, voltmeter in parallel across the component
- Components to test: ohmic resistor, filament lamp, semiconductor diode
- Switch and connecting leads

Method

1. Connect the circuit with the ammeter in series and voltmeter in parallel across the component
2. Use the variable resistor to vary the PD across the component from 0 V upwards
3. Record I and V for at least 8 values across the full range
4. For the diode: also reverse the connections to test reverse bias
5. Plot I (y-axis) against V (x-axis) for each component
6. Identify whether the component is ohmic (straight line) or non-ohmic

Safety

- Do not exceed the maximum rated current for the filament lamp
- Switch off between readings for the filament lamp to allow it to cool

Analysis

For an ohmic resistor, the I - V graph is a straight line through the origin. For a filament lamp, the curve flattens as resistance increases with temperature. For a diode, negligible current flows in

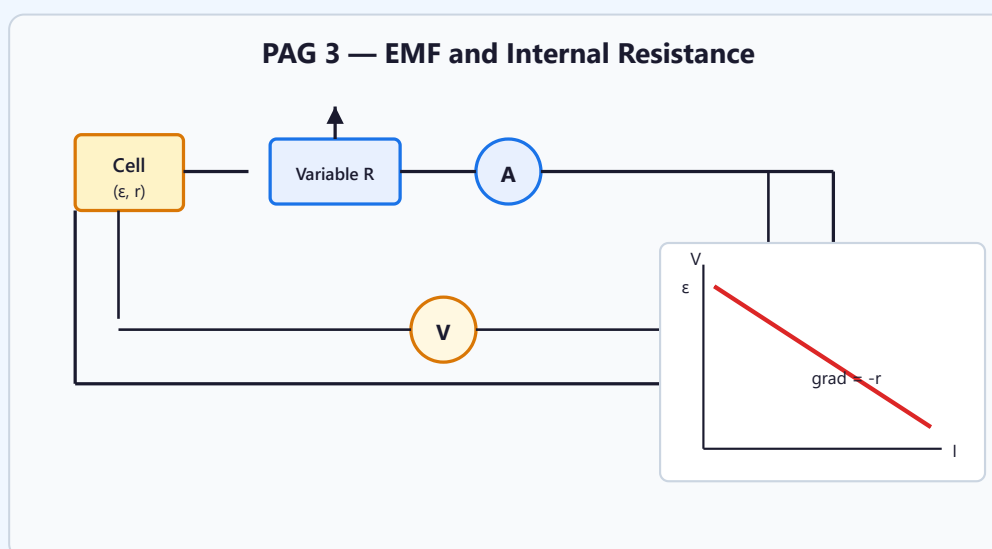
reverse bias and current increases rapidly above $\sim 0.6\text{ V}$ in forward bias.

Sources of Error & Improvements

Source of Error	Type	Improvement
Filament temperature changes during readings	Systematic	Allow time between readings for the filament to reach thermal equilibrium
Ammeter has some resistance, causing PD drop	Systematic	Use ammeter with very low resistance; account for its resistance
Reading fluctuations on meters	Random	Use digital meters; take repeat readings at each voltage

Exam Tip: Remember the difference between a rheostat (variable resistor used to vary current) and a potentiometer (used to vary PD). A potential divider circuit gives better control of PD from 0 V upward.

PAG 3: EMF and Internal Resistance



Apparatus

- Cell (e.g. AA cell, or cells in series)
- Variable resistor (rheostat) or set of fixed resistors
- Ammeter in series
- Voltmeter across the cell terminals
- Switch and connecting leads

Method

1. Connect the circuit: cell, switch, ammeter, and variable resistor in series; voltmeter across the cell terminals
2. Close the switch and adjust the variable resistor to its maximum value
3. Record V (terminal PD) and I (current)
4. Reduce the resistance to increase the current; record V and I at each setting
5. Take at least 6 pairs of readings across a range of currents
6. Plot V (y-axis) against I (x-axis)
7. The y-intercept = EMF (ϵ), the gradient = $-r$ (internal resistance)

Safety

- Do not short-circuit the cell (very low external resistance causes high current and overheating)
- Switch off between readings to preserve cell life and prevent heating

Analysis

From $\epsilon = V + Ir$, rearranging gives $V = \epsilon - Ir$. This is in the form $y = c + mx$, so plotting V vs I gives a straight line with y-intercept = ϵ and gradient = $-r$.

Sources of Error & Improvements

Source of Error	Type	Improvement
Cell runs down during the experiment (EMF decreases)	Systematic	Use fresh cells; take readings quickly; switch off between readings

Cell heats up, changing internal resistance	Systematic	Allow cooling time between readings; use low currents
Voltmeter draws small current, so V reading is slightly low	Systematic	Use a high-resistance voltmeter (digital); calculate correction if needed
Contact resistance in connections	Systematic	Ensure clean, tight connections

Exam Tip: The y-intercept gives the EMF because when $I = 0$, $V = \mathcal{E}$ (no lost volts). If the graph does not start from $I = 0$, extrapolate the line of best fit to find the y-intercept.